



Sentinel-2 Pixel Location Consistency Across Dates: Technical Foundations, Accuracy, Sources of Uncertainty, and Best Practices

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1. Introduction

The Sentinel-2 mission, operated by the European Space Agency (ESA) as part of the Copernicus Programme, delivers high-resolution multispectral imagery for global land monitoring. A foundational question for users conducting time-series analysis, change detection, or any application requiring precise spatial referencing is whether a given pixel—defined by its tile, row, column, and spatial resolution—corresponds to the exact same ground location across all acquisition dates. This question is especially pertinent for workflows that rely on extracting pixel values at fixed indices over time, such as crop monitoring, phenology studies, or surface motion analysis.

This report provides a comprehensive, technically rigorous explanation of Sentinel-2 pixel location consistency across dates for the same tile. It addresses the theoretical underpinnings of the Sentinel-2 tiling and georeferencing system, ESA's official geolocation accuracy specifications, all known technical sources of spatial misregistration and their typical magnitudes, empirical findings from peer-reviewed studies, the impact of terrain and processing improvements, and practical recommendations for robust time-series analysis. The goal is to equip users with a clear, nuanced understanding of both the strengths and limitations of Sentinel-2's spatial consistency, enabling informed decisions in scientific and operational applications.

2. Theoretical Basis: Sentinel-2 Tiling and Pixel Indexing

2.1. The MGRS/UTM Tiling System Sentinel-2 data products are organized according to the Military Grid Reference System (MGRS), which is derived from the Universal Transverse Mercator (UTM) coordinate system.

The MGRS divides the Earth's surface into a regular grid of tiles, each uniquely identified by a five-character code (e.g., 32TQM). Each tile is defined by a fixed origin and extent within the UTM/WGS84 projection. ^{1 2} ³ Sentinel-2 tiles are nominally 110 km x 110 km, slightly larger than the standard 100 km MGRS squares, to ensure overlap and avoid data gaps at tile edges. ^{3 4 5 6 7 8}

The origin and extent of each tile are fixed in UTM coordinates and do not change over time. The global MGRS/UTM grid is publicly documented and available in KML and shapefile formats, allowing users to reference the exact spatial footprint and grid origin of each tile. ^{1 2 9} This fixed grid is foundational to the spatial consistency of Sentinel-2 products.

2.2. Pixel Grid Definition and Ortho-Rectification Each Sentinel-2 tile is projected onto the fixed UTM grid, with pixel grids defined at three spatial resolutions: 10 m, 20 m, and 60 m, depending on the spectral band. The ortho-rectification process corrects for geometric distortions caused by sensor geometry, satellite motion, and terrain relief, projecting the image onto the uniform UTM grid. ^{5 6 7 8 10} The upper-left pixel of a tile always corresponds to the same UTM coordinates, regardless of acquisition date. ^{1 5 11 6}

Level-1C (L1C) products are ortho-rectified top-of-atmosphere reflectance images, while Level-2A (L2A) products are bottom-of-atmosphere reflectance images derived from L1C via atmospheric correction. The L2A processing does not alter the spatial referencing or pixel grid; it simply applies radiometric corrections. ^{4 12 13 14} Thus, both L1C and L2A products share the same fixed UTM grid and pixel alignment.

2.3. Theoretical Pixel Consistency Across Dates

Given the fixed definition of the MGRS/UTM tiling grid and the orthorectification process, the row and column indices of pixels within a tile at a given resolution are theoretically fixed to specific ground locations. For a given tile ID, the pixel at (row i , column j) at 10 m resolution should always correspond to the same UTM coordinates—and thus the same parcel of land—across all acquisition dates. ^{1 5 11 6 8}

¹⁰ This design is intentional and underpins the utility of Sentinel-2 data for pixel-level temporal analysis and change detection.

3. ESA Geolocation and Multi-Temporal Registration Accuracy Specifications

3.1. Absolute Geolocation Accuracy Absolute geolocation accuracy refers to the positional error between the image pixel and its true location on the ground. For Sentinel-2, this accuracy is achieved through precise orbit determination, sensor calibration, and geometric correction using a global reference image (GRI) and digital elevation models (DEMs).

Since August 2021, the geometric refinement process for Sentinel-2 products has relied on the Copernicus Sentinel-2 Global Reference Image (GRI), which provides a dense set of ground control points (GCPs) for geometric calibration. ^{15 16 17} The Copernicus DEM (GLO-90m, and more recently 30 m) is used for orthorectification.

The current official ESA specification is:

- **Absolute geolocation accuracy:** Better than 6 meters (at 95% circular error confidence) globally for both L1C and L2A products. ^{15 18 19 20 17}

This means that, for any given pixel, the cen-

ter is expected to be within 6 meters of its true ground position with 95% confidence. This figure is consistent across both Level-1C and Level-2A products, as geometric refinement is applied at the L1C stage and inherited by L2A. ^{21 22}

3.2. Multi-Temporal (Inter-Date) Registration Accuracy

Multi-temporal registration accuracy quantifies how well pixels from images acquired on different dates align for the same tile and resolution. This is crucial for time-series analysis, as misregistration can introduce false change signals or blur temporal trends.

The ESA specification is:

- **Multi-temporal registration accuracy:** Better than 5 meters (at 95% circular error confidence) between product pairs sharing the same relative orbit. ^{18 23 17}
- **Or, equivalently:** Better than or equal to 0.3 pixels (at 2σ confidence level) for 10 m bands, which translates to ≤ 3 meters for 10 m resolution bands. ²³

This means that, for any given pixel location (row, column) in a tile, the corresponding pixel in images from different dates will be aligned to within 5 meters (and typically much better, often within 3 meters for 10 m bands) with 95% confidence. This sub-pixel registration is achieved for both L1C and L2A products. ^{21 22}

3.3. Pre-2021 vs. Post-2021 Distinction

The deployment of the GRI-based geometric refinement in August 2021 marked a significant improvement in both absolute and multi-temporal registration accuracy. Prior to this, registration accuracy was lower, with multi-temporal errors often exceeding 12 meters (CE90) in unrefined products. ^{15 18 24} Users working with pre-2021 data should consult historical Data Quality Reports for specifics.

4. Technical Sources of Residual Spatial Misregistration

Despite the theoretical design for pixel-level consistency, several technical factors introduce residual spatial uncertainty or misregistration between images acquired on different dates, even for the same tile and nominal pixel coordinates. The main sources are:

4.1. Orthorectification Process and DEM Dependency Orthorectification corrects geometric distortions using a DEM. Any error in the DEM's representation of actual terrain elevation translates directly into horizontal misregistration. The magnitude of this error is proportional to the local terrain slope and the difference between true and modeled elevation.^{25 26} In flat areas, this effect is minimal, but in rugged terrain, misregistration can reach several meters or more. For example, a DEM vertical error of 10 m on a 20° slope can result in a horizontal displacement of approximately 3.6 m.

The Copernicus DEM's vertical accuracy is typically within 4–10 m RMSE in most regions, but local errors can be higher, especially in areas with rapid topographic change or poor DEM coverage.²⁵ The orthorectification process also involves mesh-based resampling and local linear approximations, which can introduce additional small errors, particularly at mesh boundaries or where terrain changes rapidly within a mesh cell.^{27 28}

The introduction of the GRI as a source of GCPs for geometric refinement has significantly improved geolocation accuracy and temporal consistency. Since August 2021, the use of GRI in the geometric refinement process has reduced the multi-temporal co-registration error to less than 5 m (95% CE) for refined products, compared to 12 m or more for unrefined products.^{29 30 31 32} However,

not all products are refined—cloud cover, water bodies, or isolated islands may prevent sufficient GCP extraction, and in such cases, the product geometry relies solely on the sensor model and DEM, resulting in higher uncertainty.^{33 34 35}

4.2. Sensor Attitude and Geometry Variations Sentinel-2 satellites use GPS, star trackers, and gyroscopes to determine their position and attitude (roll, pitch, yaw).^{36 37 38} Small errors in attitude determination and orbital position persist due to sensor noise, thermal deformations, or inaccuracies in the star catalog.^{39 37} The geometric model is refined using polynomial corrections based on GCPs, but residual errors remain.

The typical a priori absolute geolocation uncertainty (before GCP refinement) is about 2 km (3σ), reduced to 20 m (2σ) without GCPs and 12.5 m (2σ) with GCPs.⁴⁰ After geometric refinement with GRI, the absolute geolocation accuracy improves to 8 m (95% CE), and multi-temporal co-registration to 5 m (95% CE).^{30 31 32 41} Temporal drift in sensor alignment can introduce additional misregistration, but calibration updates are periodically applied to correct for such drifts.⁴²

4.3. Terrain-Induced Parallax Effects The Sentinel-2 MultiSpectral Instrument (MSI) employs a push-broom design with 12 staggered detectors, each covering a portion of the swath. This design introduces parallax effects due to slightly different viewing angles and acquisition times for different detectors and spectral bands.^{43 44 45 46 47 48} The maximum along-track displacement between detectors can be up to 46 km, and inter-band parallax can reach up to 14 km along track, though these are not direct pixel shifts but rather the geometric configuration of the focal plane.^{45 46}

of 1–2 m. ^{49 50}

4.6. Summary Table: Sources and Magnitude of Misregistration (Refer to Table ??)

5. Empirical Findings: Real-World Co-Registration Accuracy

5.1. Measured Pixel Shift Between Dates Empirical studies confirm that, since the adoption of the GRI in late August 2021, the inter-date (multi-temporal) co-registration accuracy of Sentinel-2 L1C products has improved substantially. The co-registration error is now estimated to be less than 6 m circular error at 90% probability (CE90) when GRI is used. ^{29 24} Prior to the GRI, errors could exceed 12 m CE90. ²⁴

A comprehensive analysis of Sentinel-2A L1C data over southern Africa found that a 2016 software update reduced the mean multi-temporal misregistration from about 1.6 pixels (16 m at 10 m resolution) to 0.4 pixels (4 m), with a standard deviation of about 0.2 pixels (2 m). ⁵⁹ More recent global assessments report that, with GRI-based processing, the temporal co-registration error between Sentinel-2 images is typically better than 5.1 m CE90. ⁶⁰

- **Absolute Geolocation Accuracy:** Close to 11 m at 95% confidence level for both Sentinel-2A and Sentinel-2B, with temporal co-registration error within one pixel (10 m) for 92% of Sentinel-2A and 87% of Sentinel-2B products. ⁶¹
- **Multi-Temporal Misregistration:** After the 2016 processing update, mean misregistration dropped to 0.4 pixels (4 m) with a standard deviation of 0.2 pixels (2 m). ⁵⁹
- **Harmonized Products:** The Harmonized Landsat and Sentinel-2 (HLS) product, which includes additional co-registration steps, achieves a global average registra-

tion accuracy of less than 10 m CE90. ⁶² For the S10 product (Sentinel-2 at 10 m), the computed circular error at 90% is 6.2 m. ⁶³

5.2. Terrain Effects: Flat vs. Mountainous Regions

Terrain-induced errors are a well-known challenge in satellite image orthorectification. In flat areas, the DEM-based correction is straightforward, and co-registration accuracy is typically at its best. In contrast, hilly or mountainous regions introduce additional sources of error:

- **Mountainous Terrain:** Studies focusing on alpine and mountainous regions report that co-registration errors can increase to 0.5–1.0 pixels (5–10 m at 10 m resolution) or more, particularly in areas with highly variable terrain relief. ^{64 65} For example, in the French Alps, analysis of 38 Sentinel-2 images found that displacement measurements were less reliable in steep terrain, with errors sometimes exceeding one pixel. ⁶⁶
- **Flat Terrain:** In flat agricultural or urban areas, co-registration errors are typically at the lower end of the reported range (sub-pixel to 0.5 pixels), with most pixels remaining well-aligned across dates. ⁵⁹

Users working in mountainous regions are advised to be cautious when interpreting pixel-level changes, as local misregistration can be higher than the global average. Some studies recommend using higher-resolution or updated DEMs, or applying additional local co-registration corrections, to mitigate these effects. ^{64 65}

5.3. Raw L1C/L2A Pixels vs. Additional Co-Registration Corrections

- **Without Additional Correction:** Raw L1C/L2A products processed with GRI typically achieve 4–6 m CE90 inter-date alignment. ^{29 60}

- **With Additional Correction:** Applying image-to-image co-registration can reduce residual shifts to 1–2 m or less, especially in flat terrain. ^{67 66} In mountainous areas, improvements are possible but may be limited by DEM quality and local relief effects.

Automated algorithms (e.g., AROSICS, phase correlation, feature-based matching) can align images to a reference date, reducing residual shifts to sub-pixel levels (often <2 m at 10 m resolution). ^{67 66} The HLS product applies additional spatial normalization and bundle adjustment, achieving global average registration accuracy of <10 m CE90, with S10 products at 6.2 m CE90. ^{62 63}

5.4. Empirical Data Visualization The following charts summarize key empirical findings:

6. Practical Recommendations for Time-Series Analysis Workflows

6.1. Quality Filtering Always apply cloud, shadow, and snow masking to remove contaminated pixels. Use the Scene Classification Layer (SCL) and, if possible, enhanced algorithms for challenging conditions (e.g., high altitudes, complex cloud cover). ^{68 69}

6.2. Temporal Compositing and Smoothing To mitigate residual misregistration and data gaps, use temporal compositing (e.g., median, maximum NDVI) or time-series smoothing/interpolation (e.g., harmonic models, splines, RBF filtering). ⁷⁰

6.3. Spatial Aggregation For field-scale or plot-level analysis, aggregate pixel values (e.g., median or mean within a plot) to reduce the impact of pixel-level misregistration. ^{71 72}

6.4. Co-Registration Correction For pixel-level change detection or when working

in mountainous terrain, apply additional co-registration using automated tools (e.g., AROSICS, phase correlation). ⁶⁷ For applications requiring sub-pixel precision, such as surface motion studies or fine-scale phenology, this step is essential.

6.5. DEM Updates Use the most accurate and up-to-date DEM available for orthorectification, especially in areas with significant topographic relief. ⁷³

6.6. Workflow Examples

- **Phenology and Vegetation Monitoring:** Co-register Sentinel-2 images with monthly Landsat composites to enhance geometric consistency, apply temporal smoothing, and aggregate values at the plot level. ^{74 75}
- **Crop Yield Estimation:** Emphasize cloud filtering, time-series smoothing, and spatial aggregation to mitigate the effects of residual misregistration. ^{76 77}
- **Change Detection and Surface Motion:** For high geometric precision, apply additional co-registration and validate alignment using ground control points or stable features. ^{66 65}

7. Conclusion

Sentinel-2's MGRS/UTM tiling system and ortho-rectification framework are designed to provide a fixed, globally consistent spatial reference for all data products. The pixel grid within each tile is theoretically constant across all acquisition dates, meaning that a given row and column at a specified resolution should always correspond to the same ground location. This design underpins the utility of Sentinel-2 data for time-series analysis and change detection.

However, due to technical realities—primarily DEM dependency, sensor attitude and geometry variations, terrain-induced parallax,

atmospheric refraction, and inter-satellite differences—there is always some residual spatial uncertainty or misregistration between acquisition dates. For products processed with geometric refinement (the default for most global products since August 2021), the typical multi-temporal co-registration error is approximately 4–6 meters (CE90), which is sub-pixel for 10 m bands but not zero. In mountainous terrain or in older, unrefined products, errors can reach 10–12 meters or more.

Empirical studies confirm that, in flat terrain, residual shifts are typically sub-pixel, while in hilly or mountainous regions, errors can increase to 0.5–1.0 pixels or more, depending on DEM quality and local relief. Raw L1C/L2A products are sufficient for most applications, but additional co-registration corrections can further reduce misregistration to 1–2 m, especially in flat areas.

For most practical purposes, extracting a pixel at a given row and column in a Sentinel-2 tile across all dates will reference the same parcel of land, with only minor spatial uncertainty due to residual geometric errors. For applications requiring sub-pixel precision or involving very small features, users should account for this uncertainty and, if necessary, apply additional co-registration or correction techniques. The ongoing improvements in DEM quality, geometric refinement algorithms, and calibration between Sentinel-2A and 2B continue to reduce these errors, but perfect spatial consistency at the sub-pixel level remains technically unattainable with current processing chains.

In summary, while Sentinel-2 provides some of the best spatial registration available in open-access Earth observation data, a small but non-negligible spatial uncertainty persists. Users should be aware that, while

row/column indices in a Sentinel-2 tile generally correspond to the same ground parcel across dates, there is always some spatial uncertainty, and best practices should be followed to minimize its impact on analysis outcomes.

Table 1: {#tbl:table_1}

Source of Misregistration	Typical Magnitude (Refined Products)	Typical Magnitude (Unrefined Products)	Notes
Orthorectification/DEM dependency	<5 m (95% CE)	8–12 m (95% CE)	Higher in mountainous terrain or with DEM errors 29 41 49 57 58
Sensor attitude/geometry variations	<5 m (95% CE)	8–12 m (95% CE)	Controlled by GCPs and geometric refinement
Terrain-induced parallax	<3 m (3 σ , inter-band)	Up to 5 m (rare, steep terrain)	Most significant in steep terrain, mitigated by orthorectification 40 49 50
Atmospheric refraction	<1 m (nadir), up to 2–3 m (edge)	<2–3 m	Minor contributor, included in geometric model 53

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